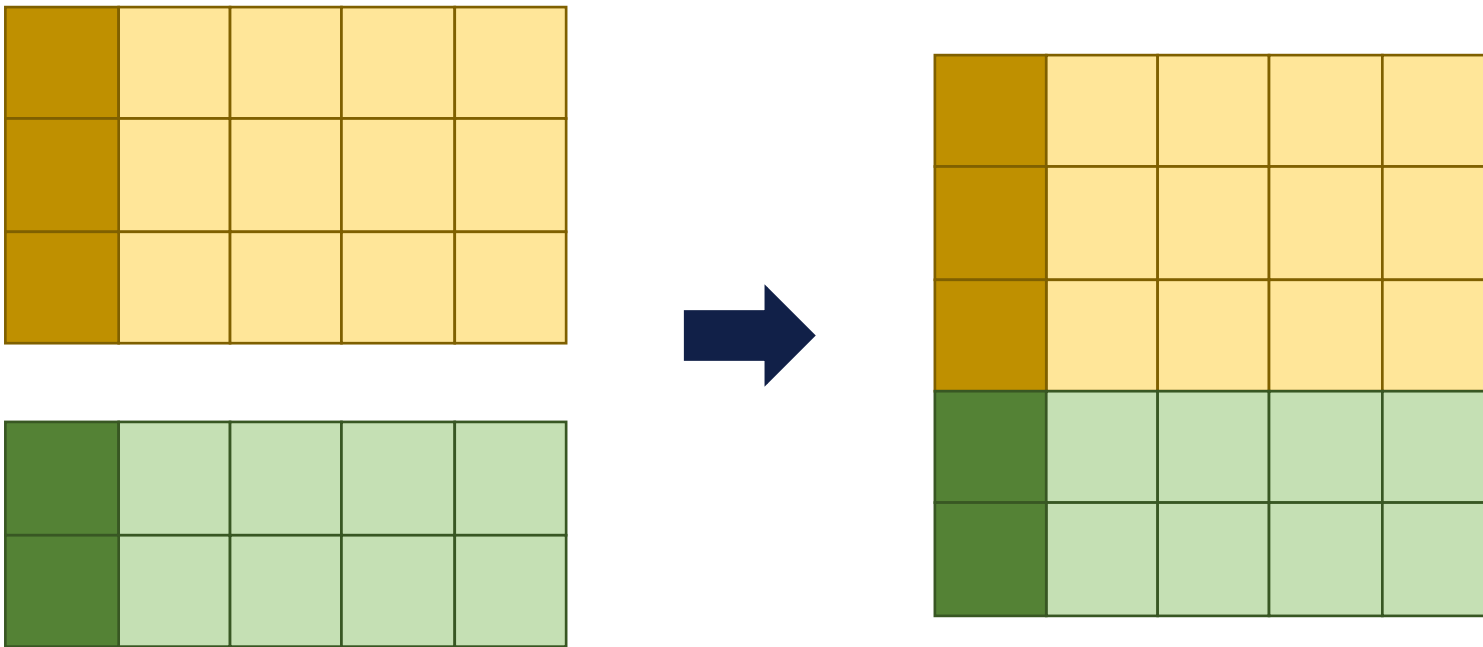


Concatenate

What if you had multiple files of the same format?

You can **concatenate** (combine) them!



Let's dissect some code...

"Concatenate the rows of data from student_df_1 with the data in students_df_2."

```
1 student_df <- student_df_1 %>%  
2   bind_rows(students_df_2)
```

Let's dissect some code...

"Concatenate the rows of data from student_df_1 with the data in students_df_2."

```
1 student_df <- student_df_1 %>%  
2   bind_rows(students_df_2)
```

bind_rows()

Specify tibbles to combine (must have the same columns!)

Demo

`bind_rows()`

Combine two tibbles

Knowledge Check

Concatenate

True or False:

You should use `bind_rows()` to combine tibbles with different columns.

Knowledge Check

Concatenate

True or False:

You should use `bind_rows()` to combine tibbles with different columns.

Must have the same columns!

Joins

What if you had to connect different data?

Student ID	Student	Student Grade Level
1	Jacob Z.	11
2	Tykesiah L.	10
3	Haneef J.	10

Assignment ID	Assignment	Max Score
1	Apple Counting	10
2	Orange Slicing	18
3	Grape Crushing	30

Assignment ID	Student ID	Score
1	1	8
1	2	9
1	3	10
2	1	16
2	2	18
2	3	17
3	1	25
3	2	29
3	3	30

You can **join** (merge) them!

Assignment	Student	Student Grade Level	Score	Max Score
1	Jacob Z.	11	8	10
1	Tykesiah K.	9	9	10
1	Haneef J.	10	10	10
2	Jacob Z.	11	16	20
2	Tykesiah L.	9	18	20
2	Haneef J.	10	17	20
3	Jacob Z.	11	25	30
3	Tykesiah L.	9	29	30
3	Haneef H.	10	30	30

Let's dissect some code...

"Join students with their scores by student IDs."

```
1 student_df %>%  
2   left_join(assignment_students_df, by = "student_id")
```

Let's dissect some code...

"Join students with their scores **by student IDs**."

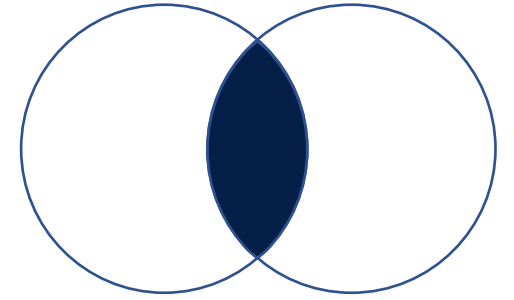
```
1 student_df %>%  
2   left_join(assignment_students_df, by = "student_id")
```

left_join()

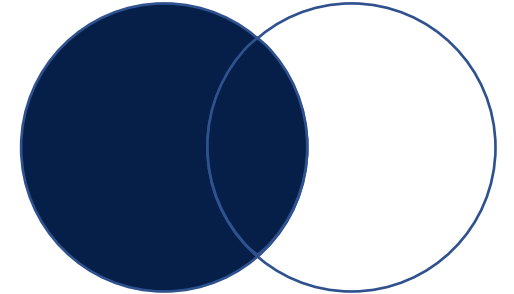
Specify which column/variable connects the tibbles.

Type of Joins

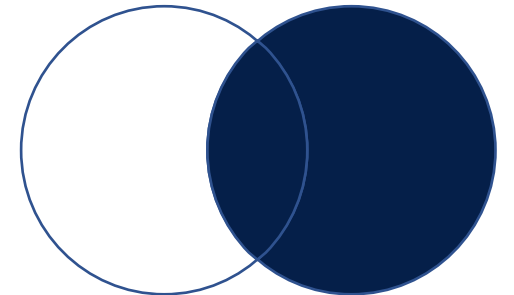
INNER JOIN



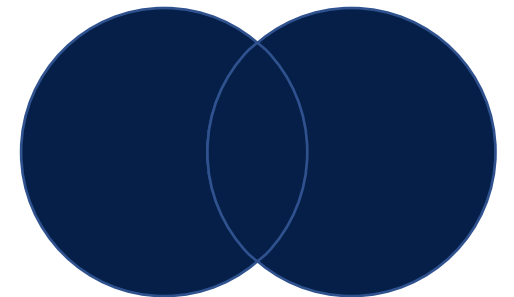
LEFT JOIN



RIGHT JOIN



OUTER JOIN



Demo

Joins

Join tibbles.

Knowledge Check

Joins

True or False:

To join two tibbles by *student_id*, both tibbles must contain the same *student_id* values.

Knowledge Check

Joins

True or False:

To join two tibbles by *student_id*, both tibbles must contain the same *student_id* values.



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